

Lecture 40: Working Up the Alternative Stack

World models, planning, and a billion-dollar bet

Every number in this deck was checked on 8 August 2026, and where a claim could not be traced to a paper it is labelled on the slide.

Last week's robots. This week's verdict on them.

Chapter 15 ended with vision-language-action models doing real work: folding laundry, clearing tables, taking out the trash.

Asked whether JEPA-based approaches would overtake them, Yann LeCun's answer was three words:

"VLAs are doomed."

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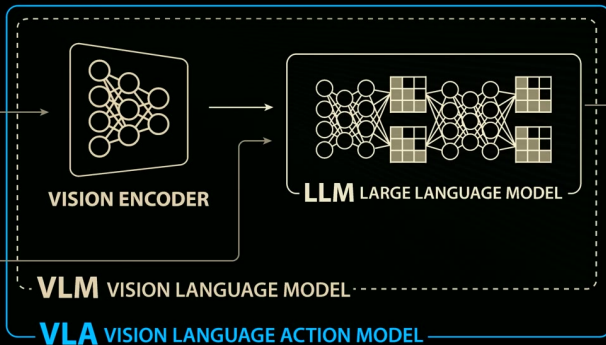
"VLAs are doomed."

Meanwhile, the JEPA-based alternative he is backing takes about **a minute** to move a cup off a platform.

Hold both of those in your head. Today is about whether the confidence is earned.



"Peel the zucchini"



Control Signals

How many hours of robot data?

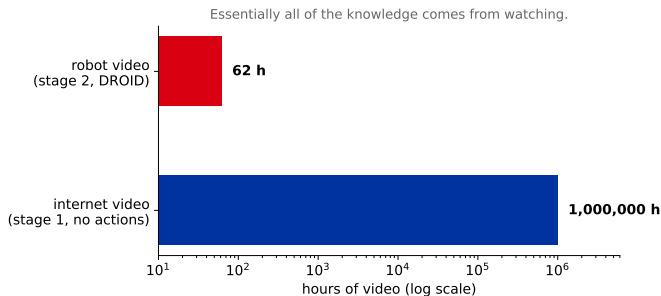
A robot picks and places objects it has never seen, in a laboratory whose data the model was never trained on, with **no task-specific training and no reward function**.

Guess the size of its robot training set.

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Guess the size of its robot training set.



Under **62 hours** of robot video - beside **a million hours** of ordinary internet video. Four orders of magnitude.

The claim

Physics can be learned by watching.

Only the mapping from *commands* to *consequences* needs a robot.

If that is true, it dissolves chapter 15's central problem. There is no internet of robot data and there never will be - but there is an internet of **video**, and video is full of objects falling, rolling, colliding and being picked up.

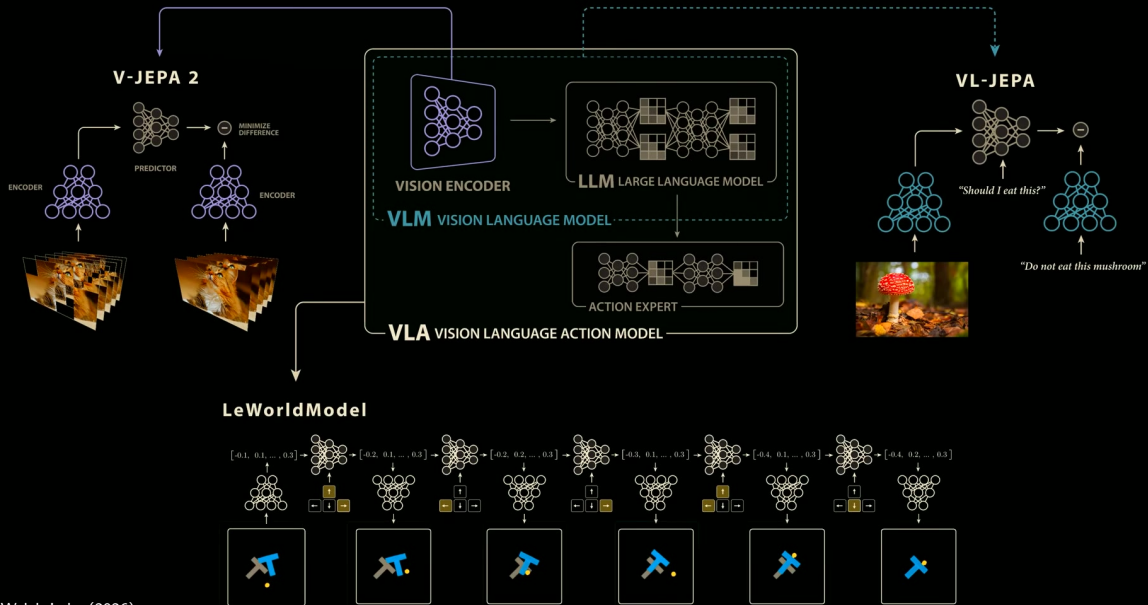
Chapter 15's answer to data scarcity was to pool across robots and squeeze more from every demonstration. This is a different answer: **stop needing the demonstrations for the physics.**

The map for today

A VLA is built on a vision-language model. A vision-language model is built on a vision encoder. **JEPA has a candidate at all three levels.**

	Mainstream	JEPA alternative
Level 3	VLA (ch15)	LeWorldModel, V-JEPA 2-AC
Level 2	vision-language model (ch12)	VL-JEPA
Level 1	CLIP-style vision encoder (ch12)	V-JEPA 2

We climb it from the bottom. **Watch the evidence as we go** - it does not stay the same strength at every level, and that turns out to be the most important thing in this lecture.



Outline

Level 1: the vision encoder

Level 2: the vision-language model

Level 3: the robot, and what a world model is

Planning as search in embedding space

The honest scoreboard, and the bet

Level 1: the vision encoder

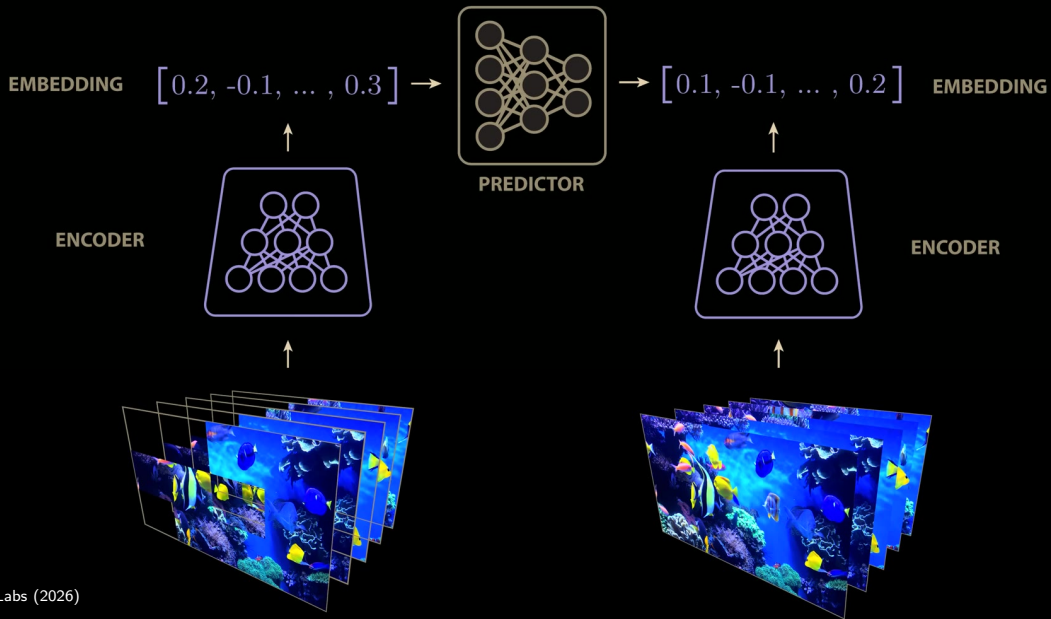
Start at the bottom. Replace CLIP.

V-JEPA: mask in space *and* time

Last lecture's recipe, with a time axis added. Corrupt a video clip by removing **spatio-temporal** regions; predict the representations of what was removed.

The bet: by learning to fill in the missing pieces of video, the model has to learn **how things move** - and by proxy, how the world works.

This is the first system in the chapter for which "predictive" means anything like "predicts the future".



Why Something-Something v2 is the honest benchmark

Most video benchmarks can be solved by **looking at one frame**. Recognise a swimming pool, guess "swimming".

Something-Something v2 is built to defeat that. Its classes are defined by **motion**:

"pushing something from left to right" vs *"pushing something from right to left"*

Identical objects, identical scene, identical single frame. Only the direction differs.

A model that has only learned appearance scores at chance here. That is why it is the number worth quoting for a video model - and why V-JEPA 2's **77.3** is meaningful.

V-JEPA 2, and the contrast that matters

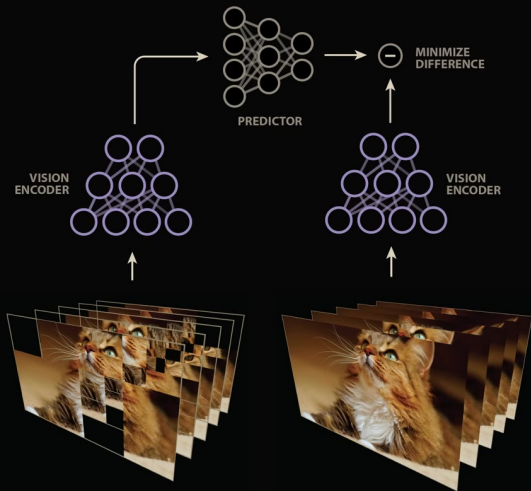
V-JEPA 2 (Meta, June 2025): **1.2B** parameters, over **one million hours** of internet video plus a million images.

CLIP is trained so image embeddings match **caption** embeddings. Its representation is **pinned to language**: it learns the distinctions that people bothered to write down.

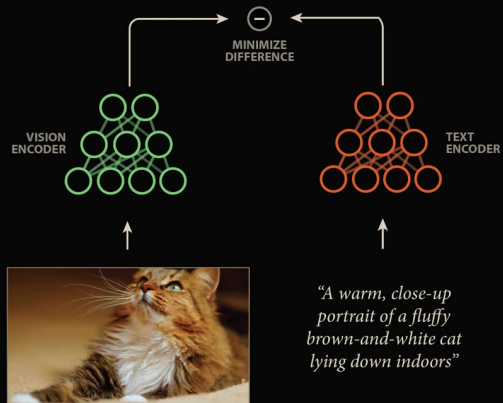
V-JEPA never sees a word. It may represent "cat" however it likes, as long as that helps fill in missing video. **Unconstrained, and unaided.**

AMI Labs - the company LeCun leaves Meta to found, and which we come back to at the end of this lecture - puts the philosophy on its landing page:
"Real intelligence does not start in language. It starts in the world."

V-JEPA



CLIP



So does a language-free encoder even talk to a language model?

It is a real worry. If nothing ever forced the representation to line up with words, why would a language model be able to read it?

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It works - and the authors flag it as "*contrary to conventional wisdom*":

Something-Something v2 (motion)	77.3
Epic-Kitchens-100 action anticipation	39.7 recall@5
PerceptionTest (with an 8B language model)	84.0
TempCompass (with an 8B language model)	76.9

Note what the last two require: the encoder is **aligned to a large language model** to produce them. At level 1 the JEPA alternative wins by **joining** the mainstream stack, not by replacing it. Hold that thought.

TEMPCOMPASS BENCHMARK

Do Video LLMs Really Understand Videos?



What is happening in the video?

- A. A person drops down the pineapple*
- B. A person pushes forward the pineapple*
- C. A person rotates the pineapple*
- D. A person picks up the pineapple*



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Level 2: the vision-language model

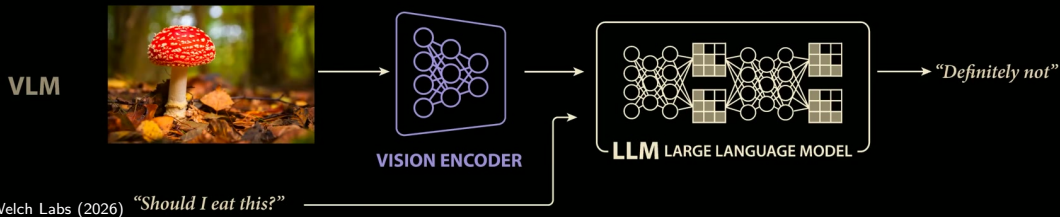
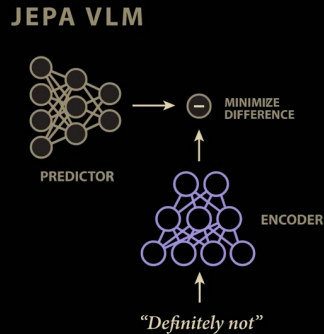
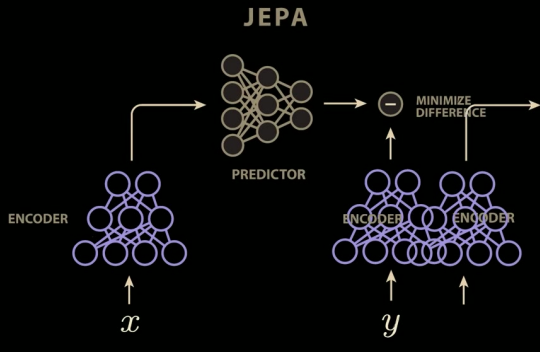
Now replace the whole thing.

The one-line change

A standard vision-language model emits **text, one token at a time**. Everything else about it - vision encoder in, prompt conditioning, a transformer in the middle - is already the shape of a JEPA.

VL-JEPA keeps all of that and changes the target: instead of generating the answer, it **predicts the embedding of the answer**, in one shot.

The prompt becomes conditioning on the predictor. The answer text goes through its own encoder to produce the target. That is the entire modification.



Why that should help: the mushroom

Show the model a mushroom and ask whether it is safe to eat. Suppose the training data says *"do not eat this mushroom"*.

Your model answers: *"this mushroom is not safe to eat."*

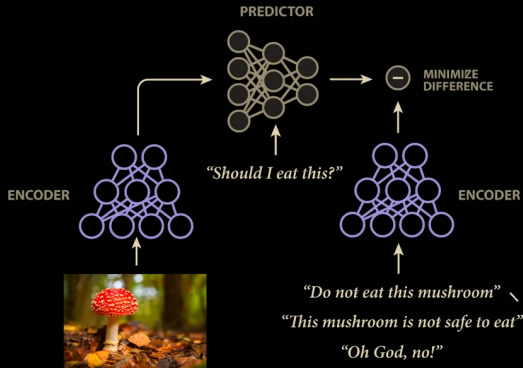
Generative VLM. Wrong tokens in the wrong order. **Penalised** - for an answer that is completely correct.

VL-JEPA. Both sentences encode to **nearly the same embedding**. The surface wording was never the target.

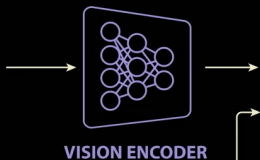
This is the leaves argument, moved from pixels to language.

Unpredictable surface detail - which leaf, which phrasing - is exactly what a learned target is free to discard.

JEPA VLM



VLM



"Do not eat this mushroom" ✓

"This mushroom is not safe to eat" ✗

"Oh God, no!" ✗

"Should I eat this?"

The controlled comparison

Same vision encoder. Same data. Same training configuration. Only the objective differs.

After 5 million training examples, zero-shot video classification:

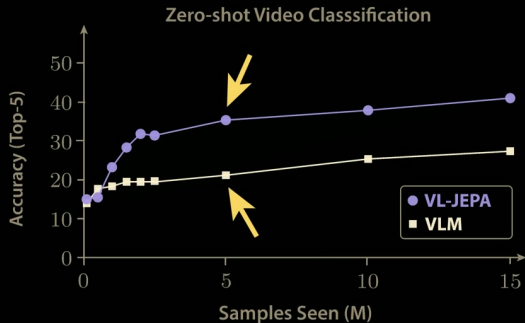
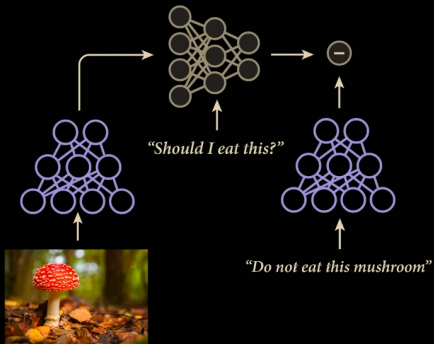
VL-JEPA 35% vs **standard VLM 20%**

And the efficiency turns into capability: roughly **1.6B** parameters outperforming **7B-13B** models on **GQA** (questions needing several reasoning steps over one image - *"is there fruit to the left of the tray the cup is on?"*), and **65.7%** on **WorldPrediction-WM** (pick what plausibly happens next) against GPT-4o's 53.3 and Gemini-2.0's 55.6.

Sourcing, since it matters: the 35/20 comparison is reported in the Welch Labs documentary, not in a paper I have read. The GQA and WorldPrediction numbers come from press coverage of the VL-JEPA paper rather than the paper itself.

VL-JEPA

Chen, DeLong, et al. "VL-jepa: Joint embedding predictive architecture for vision-language." arXiv preprint arXiv:2512.10942 (2025).



The wrinkle. Do not skip it.

VL-JEPA predicts an embedding. Embeddings are not sentences.

It cannot answer a question by default. To score those benchmarks, all candidate answers are encoded and the nearest one to the prediction is chosen - **multiple choice**. To make it speak at all, the team trained a separate **text decoder** to turn predictions back into words.

So the comparison to a 7B generative model is not quite like-for-like: one of them was asked to *write* an answer, the other to *pick* one.

This is last lecture's closing point arriving as a bill. JEPA discards what a decoder would need - so the moment you want words back, you must bolt one on.

Level 3: the robot

Top of the stack. This is where the disagreement is real.

"World model" means two incompatible things

A. Generative

Given a state and an action, produce **the observation you would see**.

Sora, Genie, Cosmos.

You can watch its predictions.

B. Latent predictive

Given a state and an action, produce **the next state in its own representation**.

JEPA.

You cannot watch it. It is judged by whether you can plan with it.

When someone says "world model", ask which one they mean. Half the disagreements in this area are people arguing past each other across this line.

The argument for B, and the honest counter

For B: rendering an imagined future commits you to detail you cannot know and do not need. Deciding whether to reach left or right does not require the texture of the tablecloth in the imagined future - and being wrong about it should not cost you.

Against B: a generative world model can be **inspected**. You can watch Sora's prediction and see the spoon pass through the bowl. A latent world model that has learned something wrong gives you **no such handle**, and debugging it is genuinely harder.

That is the price of last lecture's closing point, paid a second time: the thing that makes it efficient is the thing that makes it opaque.

New material: learning the dynamics

Chapter 11 had **model-based** reinforcement learning - but there, the model was **handed to you**: you knew $P(s'|s, a)$ and R , and planning was computation.

A line of work asks what happens when you must **learn** the dynamics instead, and learn them in a **latent space** rather than in pixels:

- ▶ **World Models** (Ha and Schmidhuber, 2018) - encode frames, learn a recurrent dynamics model, train the policy "in imagination".
- ▶ **PlaNet / Dreamer** - latent dynamics, still carrying a decoder and a reconstruction term.
- ▶ **TD-MPC** - latent dynamics with **no reconstruction at all**.

So "predict in latent space" is not new. LeCun says so himself.

What is actually new

LeCun's own framing of the lineage, which is more honest than most:

Predicting the state at $t+1$ from the state at t and an action, then optimising over action sequences to reach an outcome - *"this is classical optimal control. This is going back to the late '50s in the Soviet Union, early '60s in the West. Very classical stuff. What is not classical is you **learn** the model . . . what is even less classical is you learn a **representation** of the input that computes an abstract state, and you learn the model in that state. And that's JEPA."*

Three claims, only the third of which is new. Worth noticing how rarely a field's champion says that out loud.

The difference that explains all of last lecture

In model-based reinforcement learning, the latent state **cannot collapse** - and nobody had to invent machinery to stop it.

Why not? Because the latent is also required to predict **reward** and **value**. Those are external, grounded signals. A constant representation predicts reward terribly, so the objective rules it out for free.

JEPA has no reward. It learns from passive video with nothing external anchoring the representation.

Every anti-collapse device in L39 - the EMA teacher, the stop-gradient, VICReg, SIGReg - is **the price of dropping reward**.

Action conditioning

Two stages, and the asymmetry between them is the whole design:

1. **Stage 1, actionless.** Over a million hours of internet video. The model learns how the world moves, from watching. No actions, no labels, no rewards.
2. **Stage 2, action-conditioned.** Freeze that, and learn $s_{t+1} = P(s_t, a_t)$ from under 62 hours of unlabelled robot video with its recorded actions.

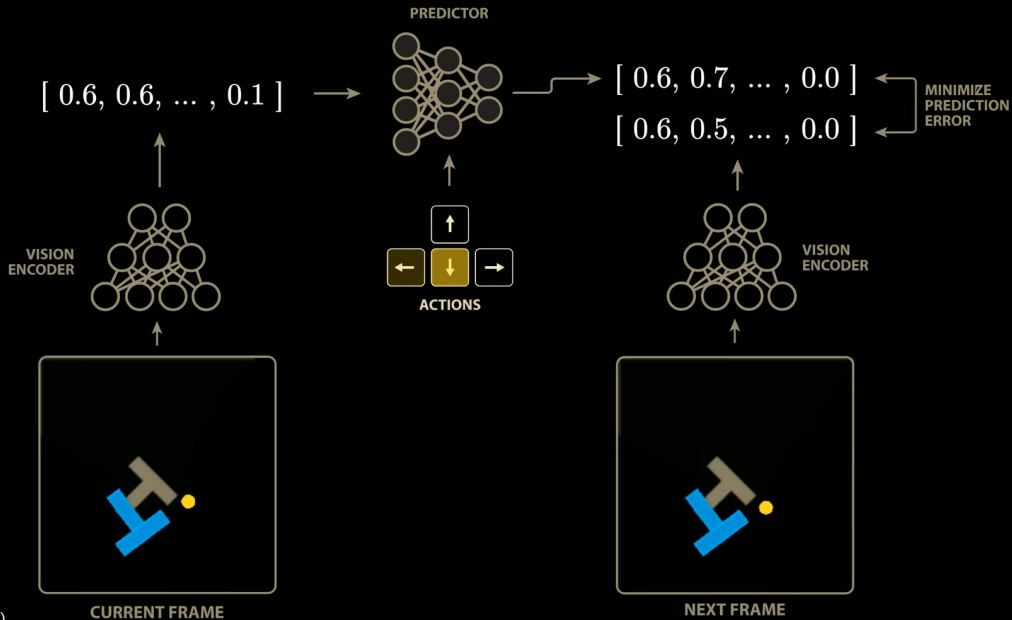
Essentially all the knowledge comes from passive observation. The robot data only has to teach it **what your commands do** - not what the world is.

Two different systems. Do not confuse them.

The rest of this lecture draws on **two** systems that share one idea. They are wildly different in scale, and nearly everything that follows depends on which is which.

	V-JEPA 2-AC (Meta)	LeWorldModel
Size	1.2B	compact; one GPU, a few hours
Trained on	1M h internet video + 62 h DROID	raw pixels and actions from one task
Runs on	real Franka arms , two labs	PushT , a 2-D pushing task
Shown here	the zero-shot result	the inside of the world model

We use the toy to **look inside** - because on PushT you can train a decoder and actually see what the model imagined. That is not possible at Meta's scale, and the pictures that follow are **not** what the Franka arm is running.



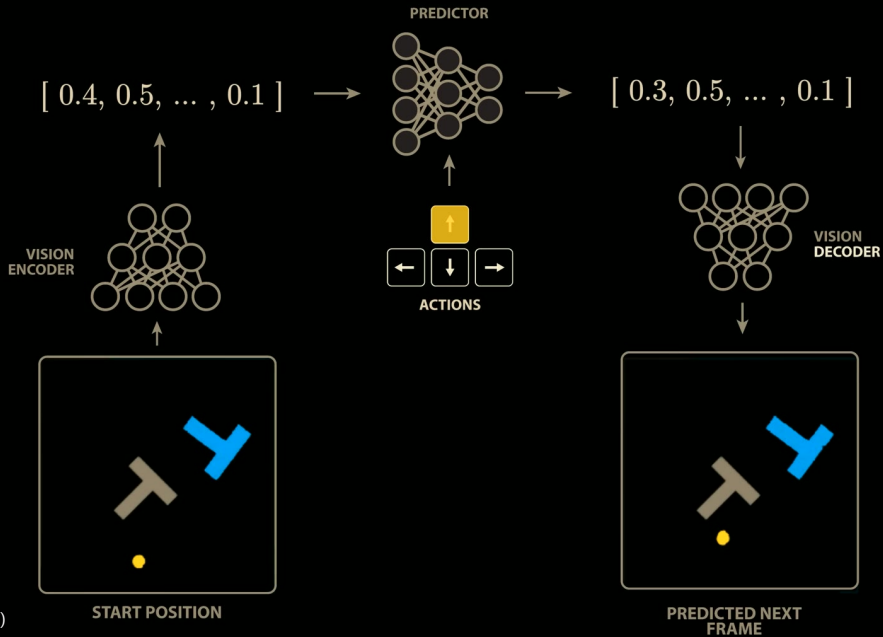
It learns the physics from pixels alone (LeWorldModel, PushT)

Train on nothing but sequences of **(image, action, next image)**. Then train a small decoder purely so we can look at the predicted embeddings.

Press *up*, and the decoded prediction moves up. Left, left. Push into the T, and the T rotates about the contact point.

Nobody told it the T is rigid, that it is movable, or how contact works. All of that is **inferred from watching pixels change under actions**.

What you have is effectively a **learned video game**: a simulator you can press keys in, built from observation instead of written by a programmer.



And it drifts (toy system: LeWorldModel on PushT)

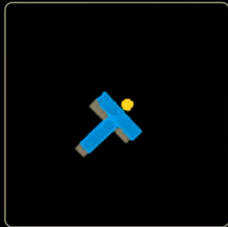
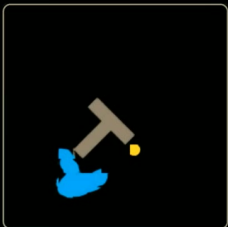
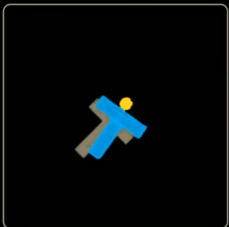
Run the model forward 18 steps and put it beside the real environment under the same actions.

Early steps match closely. Then they separate - and once they separate they do not come back, because each prediction is fed its own previous output.

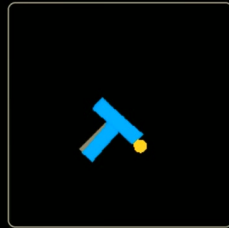
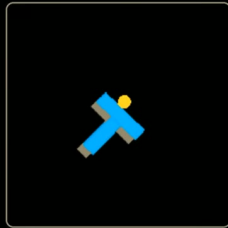
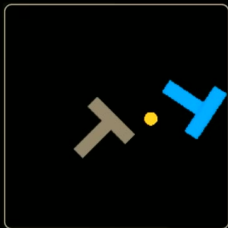
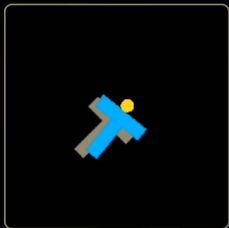
In the authors' own words, the learned world model "*goes off the rails sometimes*" - and that instability sets a hard limit on how far ahead you can plan.

Chapter 15's compounding error, third appearance: no teacher forcing at inference, so small errors feed themselves.

WORLD MODEL



REAL



Step count = 26

Step count = 28

Step count = 21

Step count = 19

Planning as search in embedding space

Now use it.

The loop

You are given a **goal image**. Encode it once; call that s_{goal} . Then, at every step:

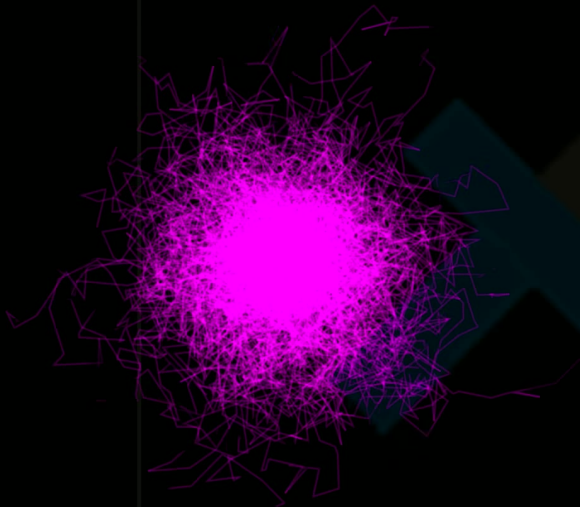
1. Sample many candidate action sequences.
2. Roll each one forward **through the world model**, in latent space.
3. Score each by $\|s_{\text{final}} - s_{\text{goal}}\|$.
4. Keep the best few, resample around them, repeat. (cross-entropy method, CEM)
5. **Execute only the first action.** Then re-plan from what actually happened. (model-predictive control, MPC)

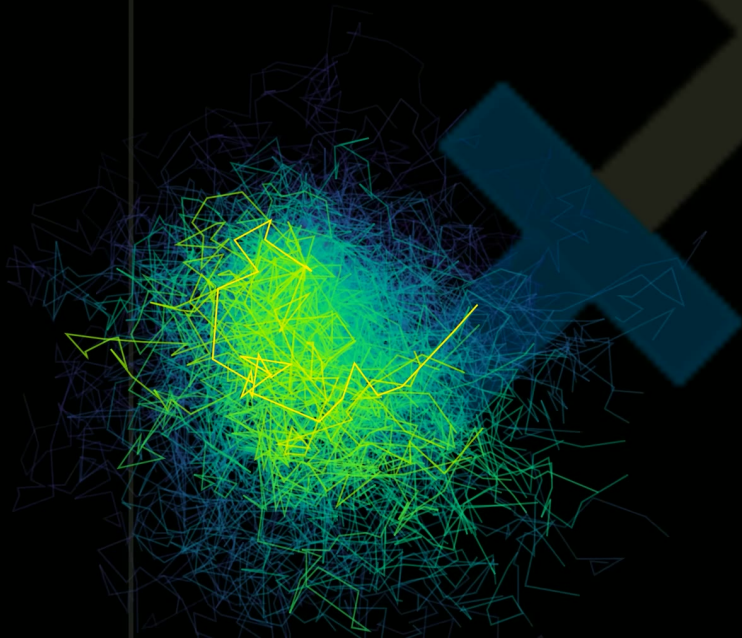
Both CEM and MPC are new here - chapter 11 planned with a *given* model, and never needed either.

The cross-entropy method, in two pictures (toy system: PushT)

Start with 500 completely random trajectories. Roll them all out in the world model, colour each by its distance to the goal embedding, keep the best 30 as an **elite set**, and resample from that set's mean and variance. Repeat.

No gradients. No trained planner. Just imagine, score, and narrow.





What the score actually is

Euclidean distance between the **predicted final embedding** and the **goal embedding**.

Notice what is *not* in that sentence:

- ▶ No reward function was specified.
- ▶ No reward model was trained.
- ▶ Nothing was ever rendered - the entire search happens in the representation, which is what makes it fast enough to run at all.

You say what you want by showing it a picture.

Execute one step, then re-plan

The rollout drifts. So do not trust it for long.

Model-predictive control plans a whole sequence, executes **one action**, throws the rest away, and re-plans from the state the world is actually in.

Chapter 15 met the same disease and chose a different cure.

Action chunking: commit to a burst of actions, so there are fewer decisions to compound.

MPC: re-plan constantly, so errors never get a long horizon to compound over.

The result, and the honest numbers (real arms: V-JEPA 2-AC)

V-JEPA 2-AC was deployed **zero-shot** on Franka arms in two laboratories Meta had collected no data from - no task-specific training, no reward - reaching **65-80%** on pick-and-place with novel objects.

Now the other side of the ledger:

- ▶ Planning takes **on the order of a minute per task** - orders of magnitude too slow for a real controller.
- ▶ **LeWorldModel** reliably plans about **five steps** ahead on PushT before the rollout stops being trustworthy.

This is not a robot controller. It is a demonstration that the **representation** contains enough physics to plan in.

The limitation hiding inside "show it a picture"

Goal-image planning is elegant. It is also the method's boundary.

This works when **the goal is known and the path is unknown.**

"Put the cup on the shelf" - fine, photograph the outcome.

"Tidy the kitchen." *"Make this safe."* *"Explore."* **There is no image to aim at.**

Open-ended tasks need exactly what this approach was advertised as avoiding: reward design, goal proposal, or something that invents intermediate subgoals.

The proposed fix: hierarchy

Two levels of world model. The **high-level** predictor proposes a **subgoal**; the **low-level** predictor plans the actions to reach it.

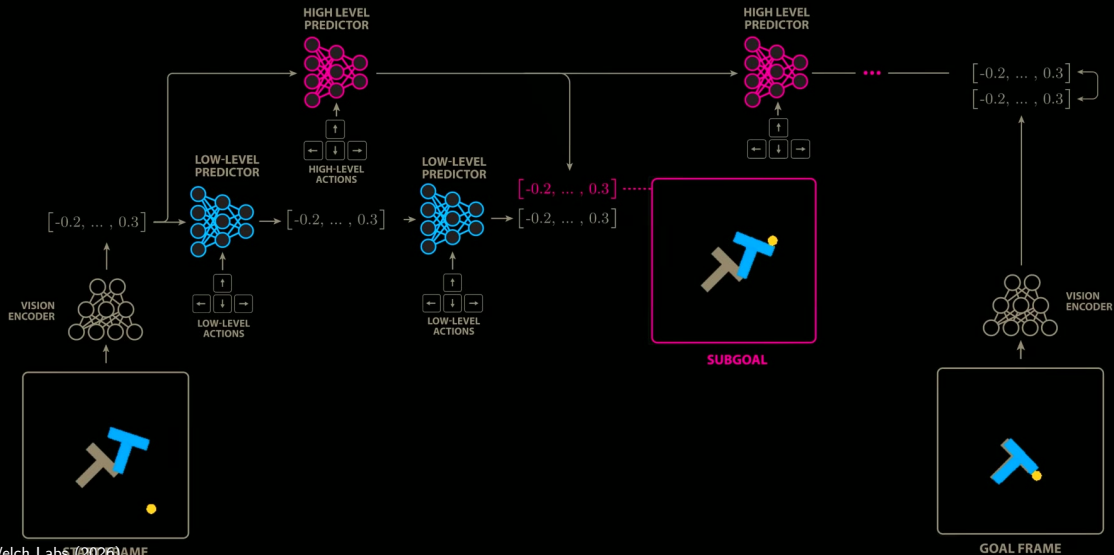
On PushT, two levels extend the reliable planning horizon from about **5 steps to 15**.

It also answers the previous slide: a hierarchy that *generates* subgoals is a mechanism for turning "*tidy the kitchen*" into a sequence of things you *can* aim a goal-conditioned planner at.

Zhang et al., *Hierarchical Planning with Latent World Models*, 2026.

Hierarchical Planning with Latent World Models

Zhang, Wancong, et al. "Hierarchical Planning with Latent World Models."
arXiv preprint arXiv:2604.03208 (2026).



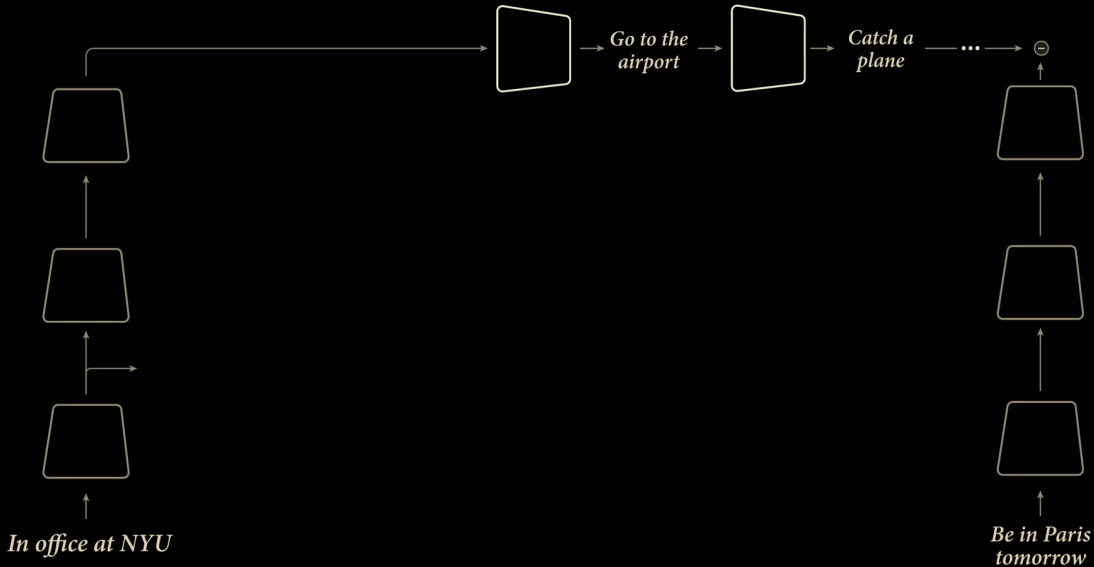
Why hierarchy is the right shape

LeCun's illustration, and it is a good one. Sitting in his office at NYU, wanting to be in Paris tomorrow:

go to the airport → catch a plane → be in Paris

He **cannot** plan that trip as a sequence of millisecond muscle commands. Not because it would be slow, but because **he does not have the information** - he does not know yet where he will be standing when he hails the taxi.

Detail buys accuracy at short range and **destroys** it at long range. So predict in detail over short horizons, and abstractly over long ones.



The open part

The hope is that the hierarchy is **learned rather than designed** - that if you train low levels to make short, detailed predictions and high levels to make long, abstract ones, a useful hierarchy of representations simply emerges, the way feature hierarchies emerged in convolutional networks.

Two honest caveats:

- ▶ LeCun concedes it requires **semi-expert trajectories** - you cannot learn high-level structure from random flailing.
- ▶ Hierarchical JEPA is still largely a **proposal**. The systems that ship are single-level.

The honest scoreboard

Meta shipped a benchmark that its own flagship model fails.

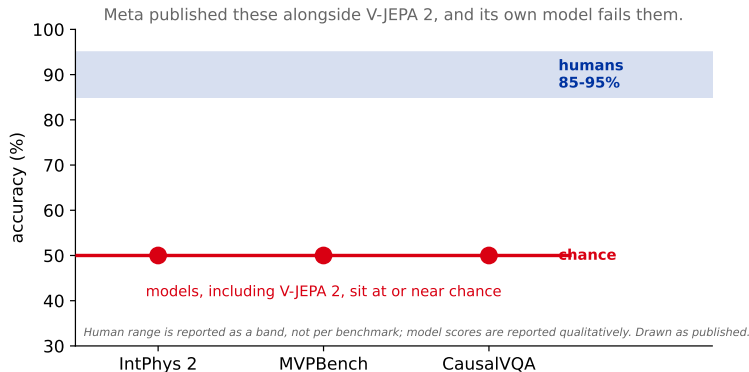
Three benchmarks, released alongside V-JEPA 2

- ▶ **IntPhys 2** - is this scene **physically possible**? Objects passing through each other, vanishing, floating.
- ▶ **MVPBench** - video-language understanding on **minimally different pairs**.
- ▶ **CausalVQA** - counterfactuals, anticipation, planning. *What **could** happen*, not what happened.

MVPBench's construction is worth stealing. Take pairs of videos that differ as little as possible but have **different correct answers**. A model that relies on a shortcut - the caption prior, a single frame - now scores **at chance by construction**.

That trick generalises far beyond video. If you build a benchmark, build it so shortcuts score zero.

The result



A model that is **state of the art at anticipating human actions** cannot reliably tell you whether a scene is **physically possible**. Whatever V-JEPA 2 learned from a million hours of video, it is **not** the intuitive physics the world-model thesis promises.

A second admission

V-JEPA 2 shipped in June 2025. In March 2026 its own successor described its representations as **"noisy"**, with **"fragmented local spatial structure"**:

	V-JEPA 2	V-JEPA 2.1
NYUv2 depth, RMSE (lower better)	0.642	0.350
ADE20K segmentation, mIoU	24.4	47.8
Something-Something v2	77.3	76.9

For nine months, the flagship world model was strong globally and close to **uninformative about where things are** - in a system whose proposed use is robot manipulation.

Note the last row: motion got slightly worse while segmentation doubled. **"The representation is good" is not one number.**

The critique of VLA, prong 1: cloning does not scale

Chapter 15's models are trained largely by **behavioural cloning** - a human teleoperates the robot, and the model learns to imitate.

LeCun's objection: you cannot collect demonstrations for every variation of every task, and outside the demonstrations these systems are, in his words, "*completely helpless ... brittle*".

A world model needs no demonstration of the task at all. It needs to know **what happens if**, and then it can search for a solution nobody showed it.



"ALOHA Robot at Stanford University" Zipeng Fu at Stanford University, mobile-aloha.github.io

Prong 2: no explicit planning

A VLA is trained and deployed **end to end**. Camera images and joint positions in, next joint positions out, repeat.

It plainly does something plan-like - chapter 15's robots pass a key between grippers to orient it correctly. But that reasoning is **implicit and invisible**.

*"If you really want to build reliable agentic systems, they absolutely have to be able to predict the consequences of their actions . . . the inference process now becomes a **search**, as opposed to just an autoregressive prediction."*

A search you can inspect, bound, and attach safety constraints to.

The counter-evidence, kept deliberately

The critique is strong. It is also not the whole picture, and leaving it unanswered would be unfair to chapter 15.

- ▶ **RT-2** moved an object to a picture of Taylor Swift. Nothing in its demonstration data involved Taylor Swift - it connected a concept from **web pretraining** to actions learned separately.
- ▶ Later systems perform tasks absent from their demonstrations: microwaves, paper towel rolls, air fryers.

Generalisation is a sliding scale, not a yes/no. These tasks were not in the demonstrations; *similar* ones were. The open question is how far that stretches - not whether it exists.



"ALOHA Robot at Stanford University" Zipeng Fu at Stanford University, mobile-aloha.github.io

The higher you climb, the thinner the evidence

Look back at the stack we climbed, and at what was actually behind each rung.

	The JEPA alternative	What the evidence actually is
Level 1	V-JEPA 2, 1.2B, shipped	Real benchmark wins - but the headline numbers come from aligning it to an 8B language model
Level 2	VL-JEPA, 1.6B, research	Beats larger models - on a multiple-choice protocol , with a text decoder bolted on to speak
Level 3	LeWorldModel / V-JEPA 2-AC	A 2-D pushing toy planning 5 steps ; a real arm at 65-80%, about a minute per task

The idea gets more ambitious as you climb. **The evidence gets thinner.**

Nobody actually has to choose

The framing all lecture has been *replace*. That is the argument as its champion states it. It is not what the evidence shows.

- ▶ V-JEPA 2's best numbers come from **pairing it with a language model**.
- ▶ VL-JEPA ships a **text decoder** so it can answer at all.
- ▶ 2026 systems couple a chapter-15 VLA to a **latent world model** and use both.

Even the documentary this lecture borrows from concludes that the JEPA approach is "*not incompatible with the current mainstream language-driven approach*" - and that on agentic and robotics problems it remains "**quite limited**".

The interesting question is not *which wins*. It is *which layer does each idea belong in*.

Score the thesis

Claim

Status, August 2026

Predicting in latent space gives good, efficient encoders

Supported

You can plan with a video world model

Supported, slowly, small scale

Watching video yields **intuitive physics**

Not supported (IntPhys 2)

It replaces next-token prediction for language

Not demonstrated

Two green, two red. Anyone who tells you this is settled - in **either** direction - is selling something.

The bet

LeCun left Meta at the end of 2025 after twelve years as its Chief AI Scientist, and co-founded **AMI Labs: \$1.03B** raised in March 2026 at a \$3.5B pre-money valuation, headquartered in Paris. **No revenue plans.**

What he says they will actually build is more specific than the coverage suggests - **not robots**:

Complex systems you **cannot reduce to equations**: a jet engine, a chemical plant, a power plant, a patient's blood sugar.

"If you can write down the equations, just write down the equations."

That is a sharper thesis than "beat the LLMs": world models are for systems where **nobody has a simulator** and data is the only route to one. Meanwhile Meta shipped V-JEPA 2.1 without him.

Recap

- ▶ Every layer of the mainstream stack has a **JEPA counterpart** - and the evidence for each gets thinner as you climb.
- ▶ A **world model** means two different things. JEPA's kind predicts **state**, not pixels: cheap and unwatchable.
- ▶ Planning is **search in embedding space** against a goal *image*. No reward is ever specified - and that is both the elegance and the boundary.
- ▶ Model-based RL avoided collapse for free because it had **reward**. JEPA pays for dropping it.
- ▶ The honest state: **good encoders, real but slow planning, no intuitive physics yet.**

Anyone who leaves this room certain either way was taught badly.
The measurements are in the lecture. The bet is still running.